

***Bacillus Licheniformis* Enzyme Mix (1 mL per 23 mg tissue):**

100 μ L b. lich 100 mg/mL (10 mg/mL final conc. - Sigma, P5380)

1 μ L 0.5 M EDTA (0.5 mM final conc. - Sigma, A8806)

899 μ L DPBS (no Ca, Mg) ThermoFisher (cat. #14190)

Preparing enzymes:

The enzyme is made up in DPBS (#14190). It is aliquoted and stored at -80 °C at 100 mg/mL in 100 μ L aliquots.

Reagents

Bovine Serum Albumin - Sigma (A8806)

Hypothermosol FRS – Sigma (H4416)

Red Blood Cell Lysing Buffer – Sigma (R7757)

Required supplies:

2 mL dounce homogenizer – Bellco (1984-10002)

Centrifuge for 1.5 mL, 15 mL conicals

Pipettes and pipet tips

15 ml Conicals (MLS)

1.5 mL tubes (MLS)

30 μ M filters - Miltenyi (130-098-458)

Petri dishes (MLS)

Razor blades (MLS)

Ice bucket w/ice

Hemocytometers - InCyto Neubauer Improved (DHC-NO1-5)

Cell yield:

~9000 cells/mg (213000 cells total from 23 mg tissue), 98% viable

Protocol:

1. Dissect out ear tissue and place in ice-cold hypothermosol on ice.
2. Using forceps, transfer ear tissue to petri dish on ice. Mince ear tissue thoroughly on ice for 3-4 min on ice into 1-mm³ pieces using razor blade while manipulating tissue with forceps – you will need to use grinding motion and vigorously break up tissue.
3. Place 23 mg minced tissue into 1 mL B. Lich enzyme cocktail. Incubate on ice.
4. Shake every min; triturate every 2 min with p1000 w/tip cut (start triturating at 2 min).
5. After 20 mins of triturating on ice, use pipet to transfer digest mix to 2 mL dounce homogenizer. Use 10 strokes of Pestle A every 2 min (4 series total, 8 min). Digest mix should become turbid.

6. At 28 min let tissue chunks settle in dounce homogenizer for 2 min. Transfer 90% of supernatant (containing released cells) using 1 mL serological pipet onto a 30 μ M filter placed on a 15 mL conical.
7. Rinse filter w/5 mL ice-cold PBS/BSA 0.04%. Save flow through on ice and keep filter on tube for 2nd layer.
8. Add additional 1 mL b. lich enzyme mix to residual tissue chunks in dounce homogenizer. Continue incubating on ice.
9. Use 10 strokes of pestle A every 2 min. for 15 additional minutes.
10. Transfer entire volume to same 30 μ M filter on 15 mL conical. Rinse with additional 5 mL ice-cold PBS/BA 0.04%.
11. Centrifuge cells for 300 g for 5 min at 4 °C.
12. Remove supernatant & re-suspend in 100 μ L ice-cold PBS/BSA 0.04% (leave in 15 mL conical).
13. Add 1 mL RBC lysis buffer to cell mix. Triturate 10X. Let sit on ice for 2 min.
14. Add 12 mL ice-cold PBS/BSA 0.04% to dilute RBC lysis buffer.
15. Spin 300 g for five min. Remove supernatant. Re-suspend in 100-200 μ L PBS/BSA 0.04%.
16. Examine using hemocytometer with trypan blue.
17. Adjust concentration to 1,000 cells/ μ L for Chromium or 100 cells/ μ L for DropSeq.



Cells from ear dissociation, 20x DIC